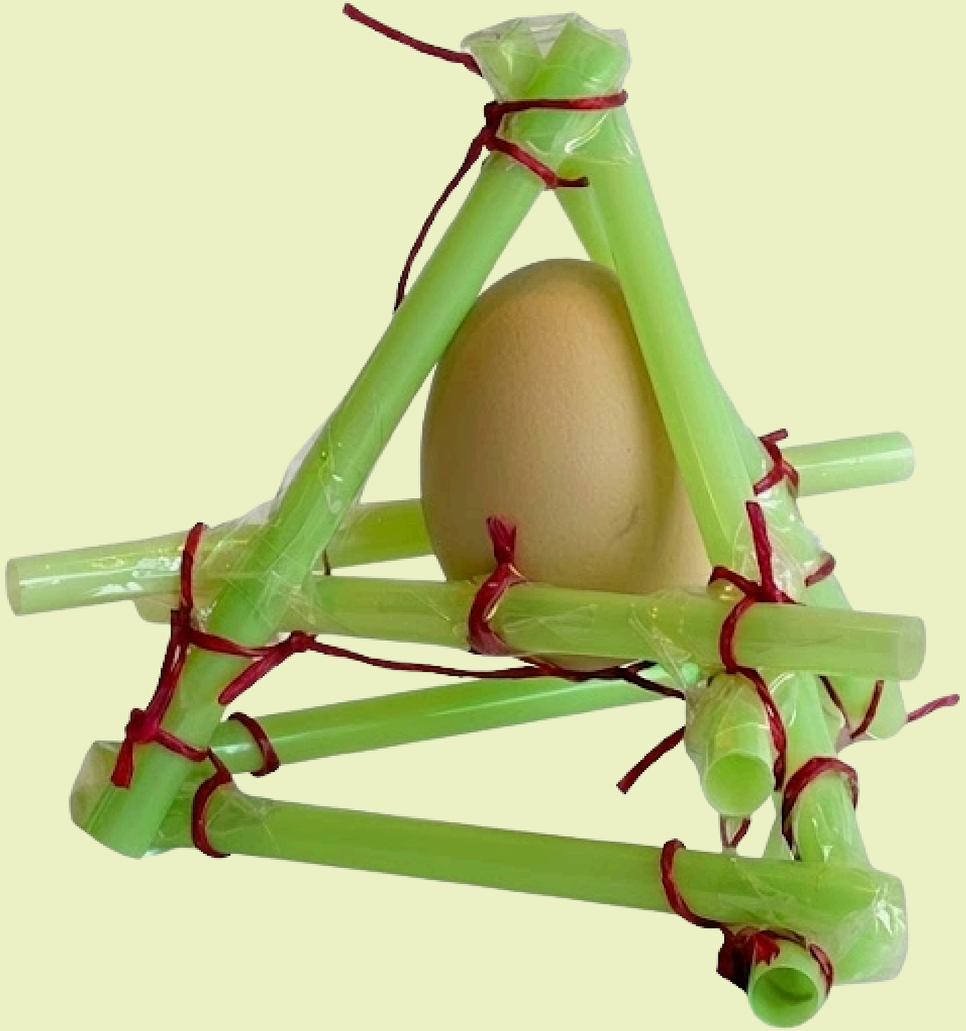
# EGG DROP CHALLENGE



IDExAI.26

DIDEM YUKSEK

# First Prototype

NO AI

**FAIL (1.5m)**

## KEYWORDS

-Octahedron
-Suspension
-Tension
-Lightweight

## KEY DESIGN MILESTONES

-M1: Initial Concept
-M2: Physical Prototyping
-M3: Evaluation/ Testing

# DESIGN DECISIONS

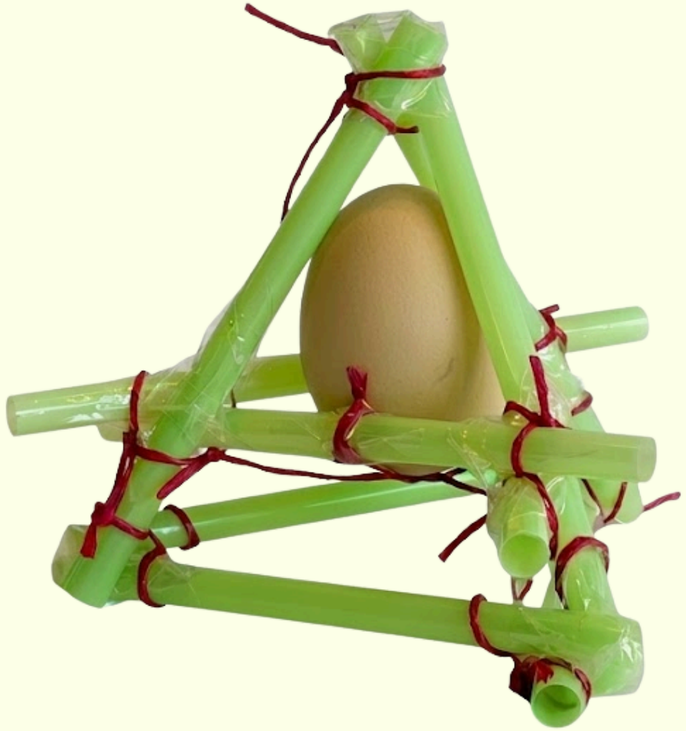
-Using an octahedron structure and placing the egg in the middle of the prototype, in order to keep it safe from crashing.

# REFLECTION

-After testing with the 1.5 meter drop challenge, I realized strings holding the egg was too loose and that’s the reason the egg broke.

# Second Prototype

WITH AI



**FAIL (3m)**

## KEYWORDS

-Recalibration
-Rigid Harness
-Impact Analysis

## KEY DESIGN MILESTONES

-M1: AI Consutation
-M2: Co Design
-M3: Physical Prototyping
-M4: Testing

# DESIGN DECISIONS

-I decided to design a straw frame with the egg suspended in the center using strings, and discussed the idea with ChatGPT received feedback and suggestions such as improving the geometry and force distribution, which helped me refine the concept.

# REFLECTION

-After the 3 meter drop, I realized the egg broke because the impact force traveled through the straw frame too quickly and the structure did not absorb enough energy, causing the egg to experience a sudden shock.

# The Role of AI: Ideation Partner, Critic

## Accepted

The shape recommendation

## Rejected

Using more tape and strings.

## Transformed

Chat GPT recommended using paper straws but I changed it with plastic ones.